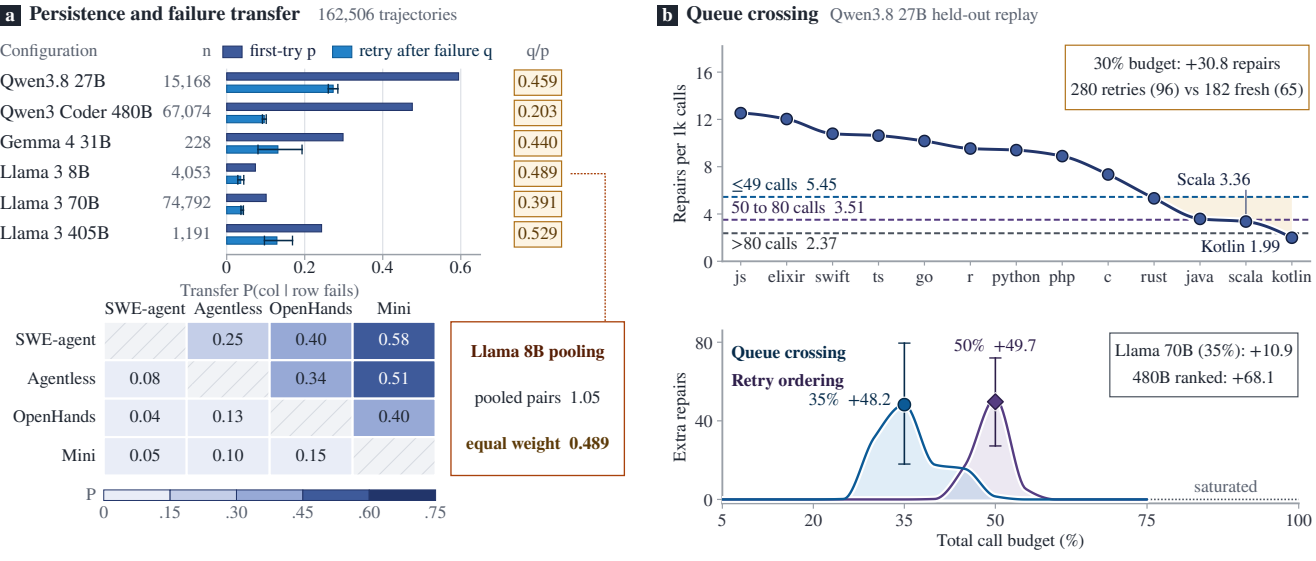
a Persistence and failure transfer 162,506 trajectories
Configuration
n
first-try p
retry after failure q
q/p
Qwen3.8 27B
15,168
0.459
Qwen3 Coder 480B
67,074
0.203
Gemma 4 31B
228
0.440
Llama 3 8B
4,053
0.489
Llama 3 70B
74,792
0.391
Llama 3 405B
1,191
0.529
0
0.2
0.4
0.6
Transfer P(col | row fails)
SWE-agent
Agentless
OpenHands
Mini
SWE-agent
0.25
0.40
0.58
Agentless
0.08
0.34
0.51
OpenHands
0.04
0.13
0.40
Mini
0.05
0.10
0.15
P
0
.15
.30
.45
.60
.75
Llama 8B pooling
pooled pairs 1.05
equal weight 0.489
b Queue crossing Qwen3.8 27B held-out replay
30% budget: +30.8 repairs
280 retries (96) vs 182 fresh (65)
Repairs per 1k calls
16
12
8
4
0
≤49 calls 5.45
50 to 80 calls 3.51
>80 calls 2.37
Scala 3.36
Kotlin 1.99
js
elixir
swift
ts
go
r
python
php
c
rust
java
scala
kotlin
Extra repairs
80
40
0
Queue crossing
Retry ordering
35% +48.2
50% +49.7
Llama 70B (35%): +10.9
480B ranked: +68.1
saturated
5
20
35
50
75
100
Total call budget (%)